

FAST ANALYTICS AND ML FOR NYC TAXI DATA

Project for scalable taxi trip analysis and
prediction

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PROBLEM STATEMENT & OBJECTIVE

Problem: Analyzing large NYC Taxi trip data to provide real-time fare predictions, duration estimates, and demand insights at scale.

Objective: Build a scalable ML and analytics system with backend REST APIs for seamless integration.

Importance: Enable accurate, fast predictions and hotspot analytics for operational efficiency and enhanced passenger experience.



TECHNOLOGY STACK

Scala – Core programming language

Apache Spark – Distributed data processing

Spark SQL – Structured queries

Spark MLlib – Machine learning library

Akka HTTP – REST API framework

sbt – Scala build tool

Docker – Containerization

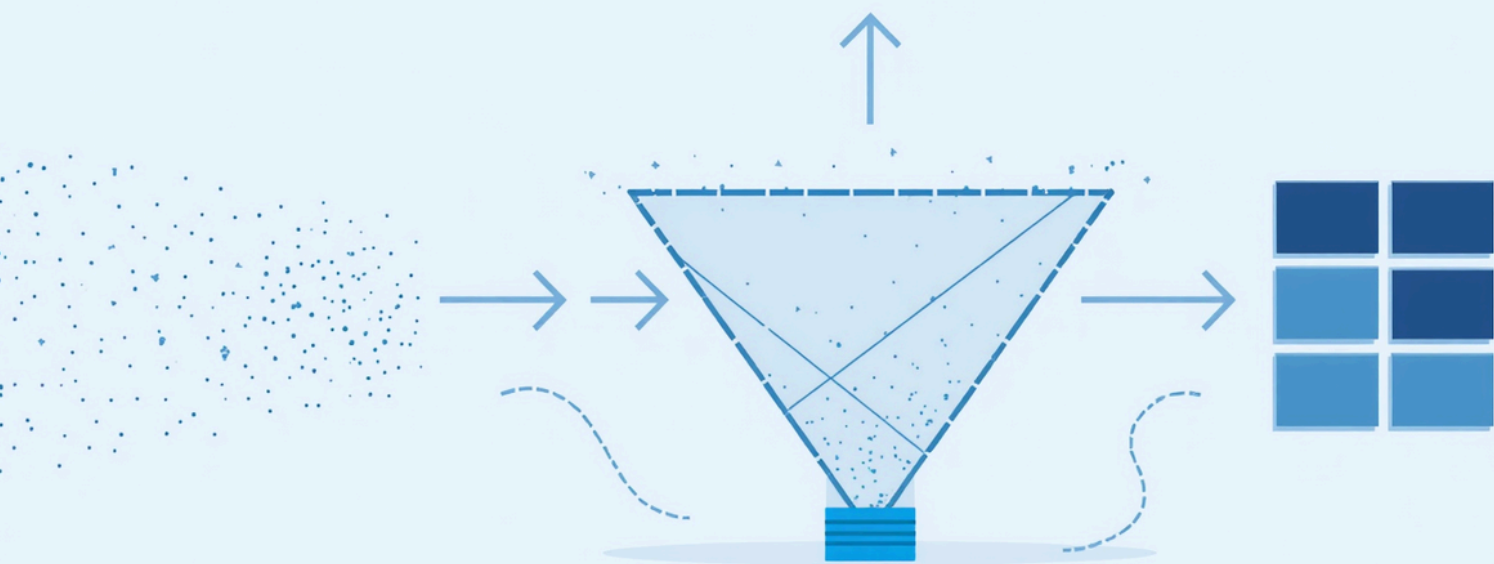


DATASET OVERVIEW

NYC Taxi Trip Records

The dataset contains millions of NYC taxi trip records with key attributes:

- Pickup/dropoff coordinates (latitude, longitude)
- Timestamps (pickup time, dropoff time)
- Fare amounts and payment details
- Trip distances in miles



Data Preprocessing

Cleaning pipeline ensures data quality:

- Remove invalid coordinates (out of NYC bounds)
- Filter negative fares and zero distances
- Handle outliers using statistical thresholds
- Address missing values in critical fields
- Result: ~95% data retention after cleaning



System Architecture

Pipeline Flow

Data Ingestion

Load NYC Taxi trip records from CSV/Parquet files into Spark for distributed processing.

Preprocessing & Feature Engineering

Clean invalid coordinates, filter outliers, handle missing data, and extract temporal and spatial features.

ML Model Training

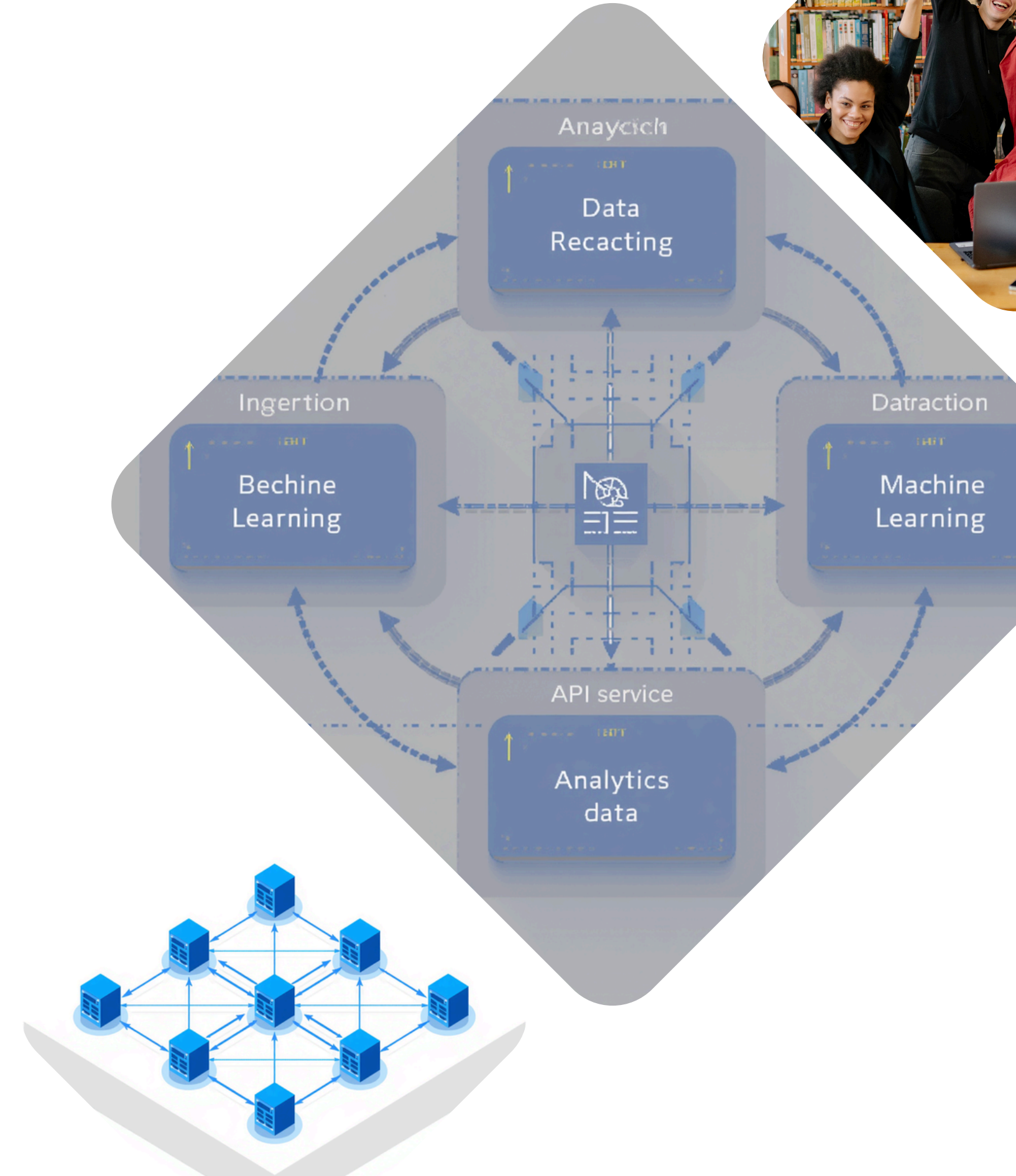
Train regression models using Spark MLlib for fare prediction and trip duration estimation.

Spark SQL Analytics

Run aggregation queries for hotspot detection and time-based trend analysis.

REST API Layer

Expose all predictions and analytics via Akka HTTP endpoints.



USE CASE 1: FARE PREDICTION

Home

Fare Test

Duration Test

Analytics

Test Fare Prediction

Submit trip features to the backend model and inspect the predicted fare.

PICKUP LONGITUDE

-73.97674560546875

PICKUP LATITUDE

40.76515197753906

DROPOFF LONGITUDE

-74.00426483154297

DROPOFF LATITUDE

40.74612808227539

TRIP DISTANCE

2.5

PASSENGER COUNT

1

PICKUP HOUR

0

PICKUP DAY OF WEEK

3

PICKUP MONTH

3

IS WEEKEND (0 OR 1)

0

Predict Fare



01

Predicts taxi fare from trip and time features including pickup/dropoff coordinates, distance, and temporal data



02

ML models trained on historical NYC Taxi trip data using Spark MLlib for scalable, accurate predictions



03

Exposed as REST API endpoint: POST /predict/fare for seamless integration with frontend applications

USE CASE 2: TRIP DURATION PREDICTION

Home

Fare Test

Duration Test

Analytics

Test Trip Duration Prediction

Submit trip features to the backend model and inspect predicted duration in minutes.

PICKUP LONGITUDE

-73.97674560546875

PICKUP LATITUDE

40.76515197753906

DROPOFF LONGITUDE

-74.00426483154297

DROPOFF LATITUDE

40.74612808227539

TRIP DISTANCE

2.5

PASSENGER COUNT

1

PICKUP HOUR

0

PICKUP DAY OF WEEK

3

PICKUP MONTH

3

IS WEEKEND (0 OR 1)

0

Predict Duration



01

Estimates trip duration in minutes before the trip starts, enabling passengers and drivers to plan accordingly



02

Uses ML models trained on historical trip data with features including pickup/dropoff coordinates, time of day, and day of week



03

Exposed as REST API endpoint: POST /predict/duration returns predicted trip time for real-time integration

DEMAND HOTSPOT ANALYTICS

Trip Analytics

Fetch and inspect hotspot, fare, and duration analytics from the backend.

HOTSPOT LIMIT

10

Load Analytics

VISIBLE HOTSPOT TRIPS

517665

PEAK AVG FARE HOUR

5

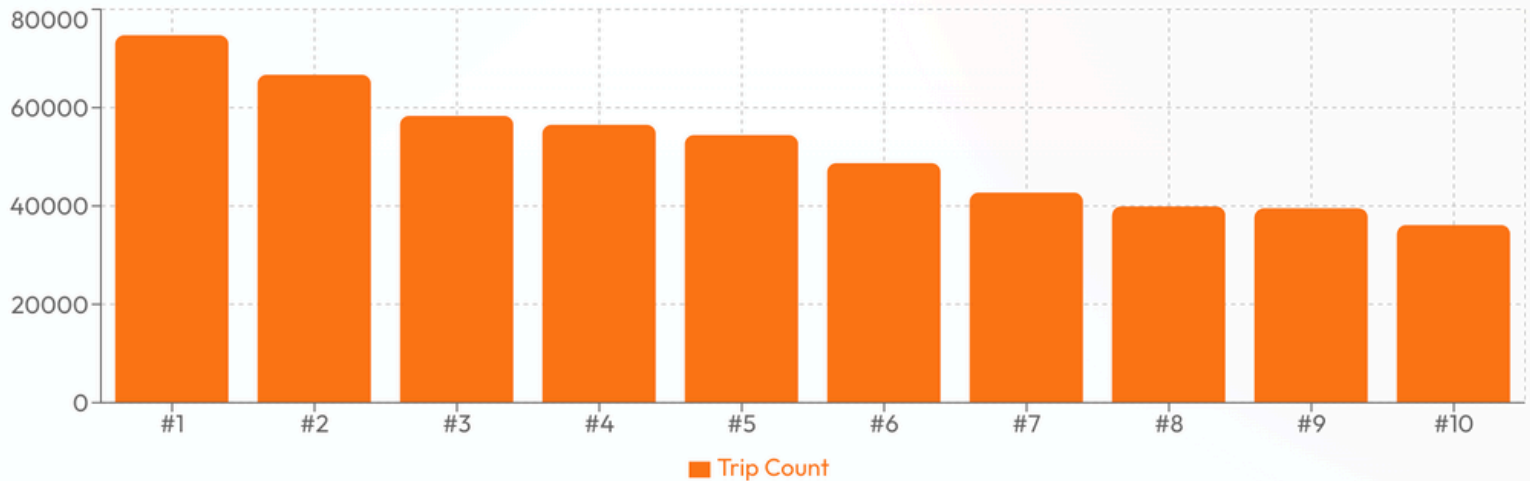
\$15.89

PEAK AVG DURATION HOUR

15

17.86 min

Top Pickup Hotspots (Bar View)



01

Identifies top pickup hotspots by trip count using Spark SQL aggregations across geographic zones



02

Analyzes average fare and trip duration patterns by hour to reveal temporal demand trends



03

REST API endpoints: GET /analytics/hotspots, GET /analytics/average-fare-by-hour, GET /analytics/average-duration-by-hour

Acceptance Criteria

Metrics:

Fare prediction accuracy: Model achieves $MAE \leq \$3.00$ on the test dataset.

Trip duration prediction accuracy: Model achieves $MAE \leq 5$ minutes on the test dataset.

API response time: 95% of API requests to /predict/fare and /predict/duration complete in under 3 seconds.

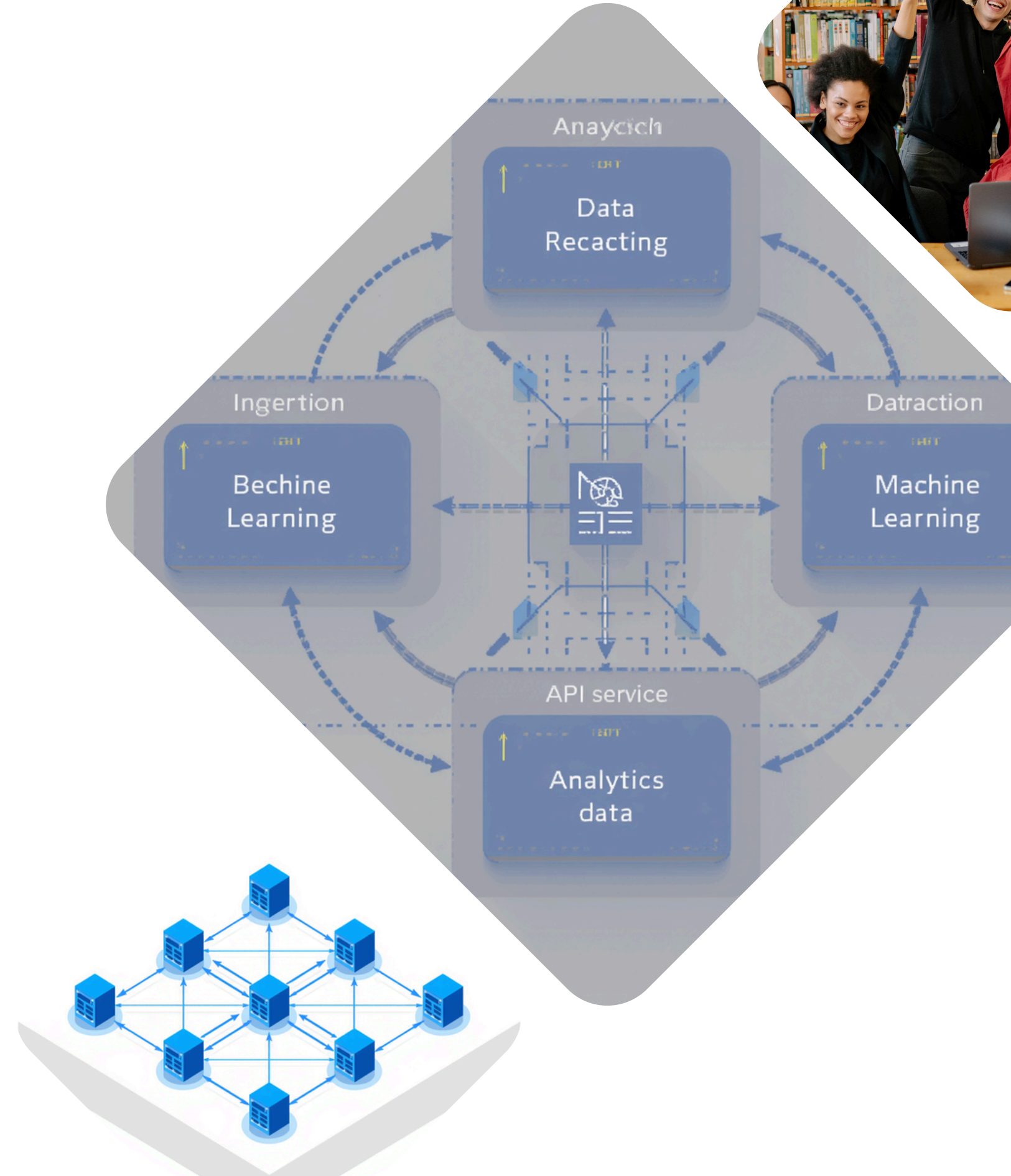
Analytics response time: 95% of requests to hotspot and hourly analytics endpoints complete in under 5 seconds.

Data quality after preprocessing: At least 90% of records retained after cleaning, while removing invalid/outlier entries.

System reliability: All core endpoints return valid JSON responses with $\geq 99\%$ success rate during demo/testing.

Scalability: System successfully processes a dataset of at least 1 million taxi trip records without failure.

Test coverage: At least 60% code coverage across core backend modules





APIS AND FUTURE FRONTEND INTEGRATION

09

POST /predict/fare

Predicts taxi fare from trip features via REST API

POST /predict/duration

Estimates trip duration in minutes before departure

GET /analytics/hotspots

Returns top pickup locations by trip count

GET /analytics/average-fare-by-hour

Analyzes fare patterns across different hours

GET /analytics/average-duration-by-hour

Provides hourly duration trends for planning

Modular, clean codebase enables easy frontend integration



THANK YOU!

Questions Welcome!



Fast Analytics and ML for NYC Taxi Data